

L04 Deep Learning Data Preprocessing

Lab thoughts / Reflective Journal

What I learned in this lab:

Completing this lab solidified my understanding that deep learning relies heavily on unspectacular work that comes before even training a model. Each data type was unique in their challenges, but the overarching concept remained consistent: raw data rarely comes in a form that's easily digestible for a model. I gained more practice resizing, normalizing, tokenizing, extracting MFCCs, and filling null values. More importantly, putting all these techniques side by side drove home the fact that all of these processes are doing the same thing—turning arbitrary, noisy input into structured data that we can expect our model to understand. I started thinking about preprocessing as an extension of modeling rather than a prerequisite.

Pitfalls I ran into and what they taught me:

There were some small struggles along the way. One part that I ran into issues with was library-specific behaviors. Some functions treated greyscale images differently (OpenCV), while other libraries hated when I did chained assignment and the model suggested updated code for pandas. While these weren't issues with the conceptual material, they forced me to slow down and pay attention to how the tools I was using expected the input data to be formatted. For my audio sample I initially received errors that I didn't realize were because there was no premade sample video loaded. When I got the basic sample to work, it came out extremely grainy compared to what I expected. Working with audio was also a challenge for me because I had to create a simple WAV file to upload into my computer for the lab itself. I also had never worked with MFCCs, which made graphing the 2D representation of a sound wave that I made a bit abstract at first. It wasn't until I actually looked at the heatmap and compared it to the sound clip I recorded that everything made sense. Each preprocessing step made me consider what assumptions I was making about the input data.

Connections to past content that stood out to me:

I felt like I had previous exposure to many of the individual techniques for preprocessing data. However, seeing how normalization and standardization applied to data beyond images helped solidify their utility for me. The same goes for tokenization when paired with

stop-word removal and lemmatization. I've also used data augmentation quite a bit in the past, but seeing how the concept applied to audio and text helped me understand it better too.

Takeaways that were totally new to me:

Something that stood out to me was how these small preprocessing steps can drastically change the makeup of our data. Rotating an image by a few degrees completely changes the distribution of pixel intensity. One missing value in a time series can alter the scale of the normalization. Preprocessing forces us to consider what information we value about the input data. I was also surprised at how much my voice affected the MFCC output. The pace I was speaking at and my deliberate tonality created a heatmap that was quite distinct from my partner's.

Applications of preprocessing that I can see myself working with:

While working through this lab, I found myself considering how each of these techniques apply to real systems. I already knew that image preprocessing is critical to many production machine learning systems, but I can also see how text preprocessing is the backbone of chatbots, search, and document classification. Audio preprocessing is at the core of voice assistants and emotion detection. I've personally worked with time series data before but saw here how preprocessing plays an important role in finance, health monitoring, and forecasting. It was interesting to make these connections and see how these individual techniques show up in the world.

Ideas I want to explore further:

I'm interested to see how preprocessing pipelines are implemented at scale. Are there automated preprocessing tools? How do you efficiently augment large datasets? How do preprocessing workflows change as we start working with transformer models? I also want to know more about how preprocessing affects fairness and bias. How do our choices in the preprocessing stage affect the rest of the pipeline? There's a lot that I still want to learn about preprocessing.